

On the Performance Impact of Poisoning Attacks on Load Forecasting in Federated Learning

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ABSTRACT

This article examines a poisoning attack on federated learning. While recent studies are actively exploring this topic in classification models of learning such as image recognition, there are few studies that address the topic in regression models. In particular, this research investigates the impacts of poisoning attacks on the performance of load forecasting, which has hardly studied yet in academia. This research implements two poisoning attacks on a federated learning setting and runs experiments to enumerate their impacts on prediction accuracy of load forecasting. With initial results, we plan to bring a couple of research questions for open discussion to audience.

CCS CONCEPTS

• Security and privacy → Distributed systems security; • Computing methodologies → Distributed artificial intelligence; • Networks → Network security.

KEYWORDS

Federated Learning; Poisoning Attack; Load Forecasting; Security; Distributed System; Energy Data; Artificial Intelligence

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1 INTRODUCTION

The prediction of energy usage in advance is known as load forecasting. We can utilize and analyze load data to understand and learn

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electric loads placed on the various systems and to find patterns in the load data to predict the future load requirements precisely. An efficient approach for load forecasting based on Long Short-Term Memory (LSTM) model has been proposed, in which electric load forecasting is done for some specific future time steps considering an hourly load demand in some specific region [2]. In contrast to image datasets such as CIFAR-10 and MNIST dataset which are the most commonly used for training various machine learning models, however, energy data is more sensitive as it contains much complicated signals and private information. Federated Learning (FL) techniques may solve this issue. Load forecasts using FL show accuracy similar to centralized models while maintaining the privacy of consumption data [3]. However, FL introduces more vulnerable surfaces due to its distributed nature. It is at an infant stage, thus such vulnerabilities and corresponding impacts on learning models have scarcely been examined yet. This article highlights a poisoning attack on FL especially and aims to investigate its impacts on the performance of load forecasting of smart energy. Using publicly available energy data, this research runs LSTM on FL for load forecasting, launches two poisoning attacks (sign-flipping attack and additive noise attack), and reports their impacts on the performance of load forecasting.

2 PRELIMINARY

Our experiments use PJM dataset [1], a regional transmission organization. It records and publishes hourly electricity consumption data i.e. hourly load supplied by electricity distribution companies. We used 11-day long dataset containing 264 hourly day records. First 10 days for training (i.e., 240 samples) and the last day for testing. In LSTM, epoch is set to 80. We have used TensorFlow for FL setup and Mean Absolute Percentage Error (MAPE) as a measure of prediction accuracy of forecasting models.

This research considers two types of poisoning attacks. In a *sign flipping attack*, first, an adversary conducts an attack by flipping the sign of the weights from positive to negative and from negative to positive. This attack differs from other data poisoning attacks as it is a quite strong attack and strongly affects the influence of untargeted clients. Second, an *additive noise attack* requires only the transferring updated weights W_c to be poisoned by adding random Gaussian noise to them. Figure 1 shows poisoning attacks against FL, in which there are (n-1) clients. Each client performs following 3 steps. (1) Each client receives an initial model from a

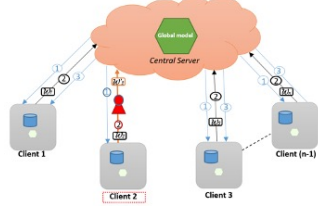


Figure 1: A poisoning attack against Federated Learning.

server. (2) It trains the initial model with its local data and sends back the updated model to the server. (3) The server aggregates all the updates received from all the clients and then sends back an updated model M to the clients. These steps are repeated to further update and improve the model's accuracy. It also shows that *Client 2* has been compromised by an attacker in the middle of a transmission. During transmission of the updated model weight W_2 in step 2, the attacker launches an attack on the weight W_2 to manipulate the results.

3 EXPERIMENTS AND RESULTS

3.1 Impact of Sign Flipping Attack

In the experiment, there are 30 clients in total and all of them participate in each round of FL. Figure 2a shows a load forecast trend with a sign flipping attack and with different settings of multiple attackers i.e. number of attackers = [0,1,2,3,4,5,6,7]. We observe a strong impact of the attack even if the number of clients is set to be 15 and the number of attackers is 2, the forecast result is almost flat as compared to the actual forecast result. The fluctuating trend of the forecasted result clearly shows that the attacker succeeds in manipulating the results and minimizing the accuracy of load forecasting. Figure 2b shows that the attacker participates in each round, flipping the sign of the weights continuously. Hence the accuracy drops tremendously with a MAPE value of 13.98%. The fluctuations in the start are due to the randomness and when the model starts to train the fluctuations become patronized. Figures 2c and 2d shows the MAPE trend without and with attack on FL. By comparing the MAPE values, we can see the impacts of the sign flipping attack without any detection algorithms in LF. Graphs show that the MAPE increases as the number of adversaries increases. In summary, the percentage error has increased from 4.582% to 14.921% as the number of adversaries has increased to 7.

3.2 Impact of Additive Noise Attack

For additive noise attack, we have set the intensity as [0.001 0.01 0.1] and we see from Figure 3a that when the noise intensity is set to 1 the forecasting result reaches infinity and the load forecasting trend for all other settings seems to be flattened and the increase/decrease trend is not visible at all. By zooming up the result we can have a visible result as shown in Figure 3b for the load forecasting. When we set the noise intensity parameter to 1 the MAPE value is extremely high. To solve this issue, we performed another simulation with a different set of noise intensity i.e. noise intensity for [0.01 0.04 0.07 0.1], and Figure 3c draws results. We observe that the trends for each setting goes similarly to each other

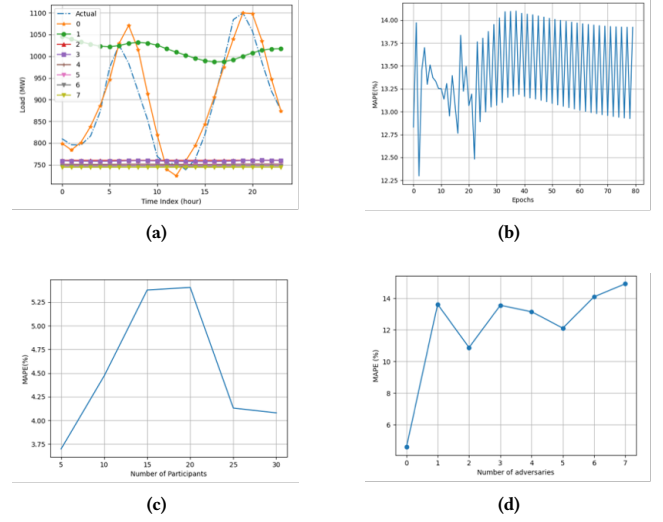


Figure 2: (a) Forecast results with a sign flipping attack and (b) MAPE trend (MAPE = 13.98% on average). Below subfigures draw the MAPE trends (c) without attacks and (d) with attacks

and no value reaches infinity, but the impact of additive noise attack depends on the intensity.

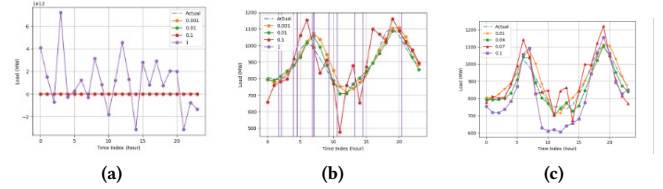


Figure 3: (a) Forecast results with an additive noise attack and (b) zoom up version. The last subfigure draws the forecast results with different noise intensities.

4 CONCLUSION

This article has enumerated the impacts of two poisoning attacks on the accuracy performance of load forecasting running on a federated learning setting. With a sign flipping attack, a prediction error increased from 4.582% to 14.921% as the number of adversaries has increased to 7 out of 30 participants. With an additive noise attack, the error increased from 3.58% to 10.72% when noise intensity was set to 0.01 and 0.1. Our experiments and results observed that load forecasting and energy data is sensitive and vulnerable to poisoning attacks and enumerated such sensitivity and vulnerability. Based on initial observation, one of future work will propose and develop an algorithm for detecting poisoning attacks in load forecasting. Open research questions for further research may include (1) other attack models for load forecasting to examine more and (2) general detection algorithms appropriate for poisoning attacks in FL.

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